

Determination of Fringe Pattern and Shape in Double Slit Experiment by Analytical Methods

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Abstract

This project investigates the formation of fringes due to interference in the double slit experiment with a screen whose inclination is a controllable parameter. The results show how the bigger picture relates to the geometry of conic sections.

1 Introduction

In this project we deal with a version of Young's Double Slit experiment where we have the screen inclination as a controllable parameter. This helps us generate fringes of different shapes and patterns. This project is theory based and we use analytical methods to determine the fringe shape and pattern.

2 Description of Our Problem

2.1 The Setup

The problem is based on the typical setup of Young's Double slit Experiment. We have parallel rays of light approaching a wall with two closely spaced parallel narrow slits where the slit width and slit separation is fixed. The slit width is of the order of the wavelength of light and slit separation is d is of the order of a millimeter. The screen is at a distance D and is inclined to the wall at an angle ψ as shown in the diagram.

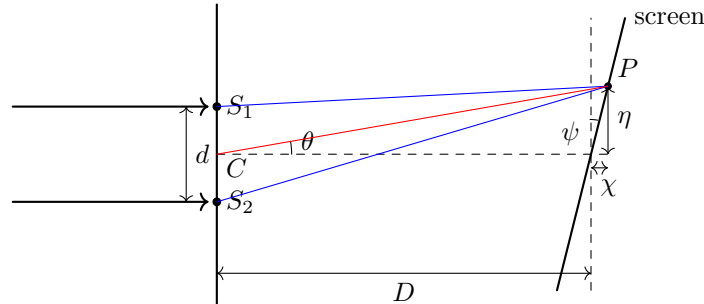


Figure 1: Young's double slit setup with slit separation d , screen distance D , and screen inclination ψ .

2.2 Derivation of the Equation of Fringes

We first try to determine the path difference expression. We know,

$$\begin{aligned}
 \Delta x &= S_2P - S_1P \\
 &= \sqrt{\left(\eta + \frac{d}{2}\right)^2 + (D + \chi)^2} - \sqrt{\left(\eta - \frac{d}{2}\right)^2 + (D + \chi)^2} \\
 &= (D + \chi) \left\{ \sqrt{1 + \left(\frac{\eta + \frac{d}{2}}{D + \chi}\right)^2} - \sqrt{1 + \left(\frac{\eta - \frac{d}{2}}{D + \chi}\right)^2} \right\} \\
 &\approx \frac{\eta d}{D + \chi}
 \end{aligned}$$

using binomial approximation. Now $\frac{\eta}{D + \chi} = \tan \theta$, so $\Delta x = d \tan \theta$

It is interesting to observe that path difference is dependent upon the θ and this proves a well known fact that the fringes are fringes of equal inclination.

Fringes are like contour plots such that they will have Δx constant along one curve. So in general if we want to determine the equation for fringes it will look something like:

$$F(x, y, \delta) = 0$$

where x, y are cartesian coordinates and

$$\delta = \frac{2\pi}{\lambda} \Delta x$$

is the phase difference, for path difference Δx and λ is the wavelength of light. It should be put to notice that x and Δx have no relation, in the sense that x is one of the cartesian coordinate on the plane where the fringes are formed and Δx is the path difference at the fringe. A fringe curve consists of all points of same path difference.

Now we set up our co-ordinate axis to properly define our cartesian coordinates. Hence we have:

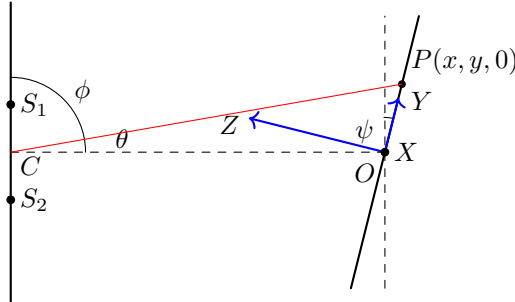


Figure 2: Setting up our three dimensional coordinate axes

We define θ and ϕ to be complementary angles, i.e., $\theta + \phi = \frac{\pi}{2}$ and this will become important very soon.

If one observes carefully one can feel an intuition that when the $\overrightarrow{CP}$ is rotated about the axis through S_2S_1 keeping ϕ constant then it generates a cone and the fringes formed are basically intersections of the cone generated with the plane of the screen. This is **the single most important idea** here and this intuition form the entire basis of the project. Let us test this! The inclination of our screen, as shown is ψ .

From the diagram,

$$\begin{aligned} O &\equiv (0, 0, 0) \\ C &\equiv (0, -D \sin \psi, D \cos \psi) \\ P &\equiv (x, y, 0) \end{aligned}$$

And hence we can write:

$$\overrightarrow{CP} = (x, y + D \sin \psi, -D \cos \psi)$$

We develop a unit vector along S_2S_1 , then

$$\hat{t} = (0, \cos \psi, \sin \psi).$$

Now,

$$\begin{aligned} \overrightarrow{CP} \cdot \hat{t} &= \|\overrightarrow{CP}\| \|\hat{t}\| \cos \phi \\ \Rightarrow (y + D \sin \psi) \cos \psi + (-D \cos \psi) \sin \psi &= \sqrt{x^2 + y^2 + D^2 + 2Dy \sin \psi} \cos \phi \\ \Rightarrow y^2 \cos^2 \psi &= x^2 \cos^2 \phi + y^2 \cos^2 \phi + D^2 \cos^2 \phi + 2Dy \sin \psi \cos^2 \phi \\ \Rightarrow x^2 + y^2 \left(1 - \frac{\cos^2 \psi}{\cos^2 \phi}\right) + 2Dy \sin \psi + D^2 &= 0 \end{aligned}$$

Now,

$$\cos \phi = \sin \theta \approx \tan \theta = \frac{\Delta x}{d} = \frac{\delta \lambda}{2\pi d}$$

So the final equation becomes:

$$x^2 + y^2 \left(1 - \frac{4\pi^2 d^2 \cos^2 \psi}{\delta^2 \lambda^2}\right) + 2Dy \sin \psi + D^2 = 0$$

For bright fringes we have $\delta = 2n\pi$ and for dark fringes we have $\delta = (2n + 1)\pi$.

If we try to plot this equation in some graphing software like Desmos, for $\delta = 2n\pi$, we get something like this:

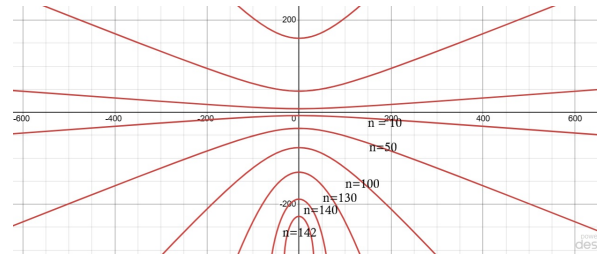


Figure 3: Fringe Shapes

3 Analysis of Equation

Our equation resembles the equation of a conic. The general form of a conic is the general second degree equation:

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

Here we have

$$\begin{aligned} a &= 1 \\ h &= 0 \\ b &= \left(1 - \frac{4\pi^2 d^2 \cos^2 \psi}{\delta^2 \lambda^2}\right) \\ g &= 0 \\ f &= D \sin \psi \\ c &= D^2 \end{aligned}$$

First we need to check whether this equation actually represents a conic. For that we check whether Δ where

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

So,

$$\begin{aligned} \Delta &= \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 - \frac{4\pi^2 d^2 \cos^2 \psi}{\delta^2 \lambda^2} & D \sin \psi \\ 0 & D \sin \psi & D^2 \end{vmatrix} = D^2 \left(1 - \frac{4\pi^2 d^2 \cos^2 \psi}{\delta^2 \lambda^2}\right) - D^2 \sin^2 \psi \\ \Delta &= D^2 \left(1 - \frac{4\pi^2 d^2 \cos^2 \psi}{\delta^2 \lambda^2}\right) - D^2 (1 - \cos^2 \psi) \\ \Delta &= D^2 \cos^2 \psi \left(1 - \frac{4\pi^2 d^2}{\delta^2 \lambda^2}\right) \end{aligned}$$

which in most of the cases is not zero (like $\psi \neq \frac{\pi}{2}$, etc.) What conic the equation shall represent is given by the factor $h^2 - ab$.

If

$$\begin{aligned} h^2 - ab > 0 &\implies \text{Hyperbola} \\ h^2 - ab = 0 &\implies \text{Parabola} \\ h^2 - ab < 0 &\implies \text{Ellipse} \end{aligned}$$

$$\text{In our equation } h^2 - ab = -\left(1 - \frac{4\pi^2 d^2 \cos^2 \psi}{\delta^2 \lambda^2}\right) = \frac{4\pi^2 d^2 \cos^2 \psi}{\delta^2 \lambda^2} - 1$$

Now,

$$\begin{aligned} h^2 - ab &= \frac{\cos^2 \psi}{\tan^2 \theta} - 1 \\ &\approx \frac{\cos^2 \psi}{\sin^2 \theta} - 1 \\ &= \frac{\cos^2 \psi}{\cos^2 \phi} - 1 \end{aligned}$$

So to get a parabolic fringe, we will need:

$$\begin{aligned} h^2 - ab &= 0 \\ \text{or, } \psi &= \phi \end{aligned}$$

Since ϕ is constrained by the small angle approximation of θ , i.e., $\theta \leq 10^\circ$, $\phi \geq 80^\circ$ and so taking $\psi \geq 80^\circ$, the parabolic (and further up elliptic) fringes will be observable. If we use light of larger wavelength then it will be easier to get within the small angle approximation range.

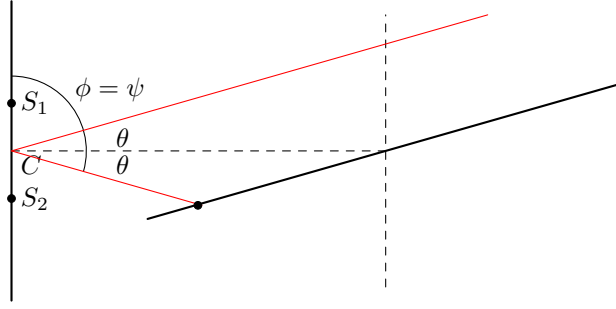


Figure 4: For Parabolic Fringe

Asymmetry: Why it should arise?

Let us note the case observed: $\psi = \phi$. The diagram looks like Figure 4.

The upper rays do not converge on the screen. Moreover for the angles of $\phi < \psi$ for the elliptic fringe range too, the same thing happens. But it does for the lower half on the screen. Since in our Δ expression, we have $\cos^2 \psi$ which is an even function and so it works as good for $-\psi$ as for ψ . Thus we have only hyperbolic fringes on the upper half while we have hyperbolic, parabolic and elliptic fringes for the lower half of the screen.

4 Experimental Validation

Although this project is primarily theoretical, experimentally observing the predicted fringes would be highly insightful. Such a setup is not currently available in our laboratories. While the apparatus itself is not particularly difficult to construct, it may not address an immediate real-world application.

That said, engaging with such experiments holds intrinsic value. Physics is not only about solving practical problems, but also about uncovering patterns, structures, and underlying principles. Work that appears purely conceptual today can become foundational tomorrow—what begins as intellectual curiosity may eventually shape transformative technologies.

5 Python Simulation

We have developed simulation for this project using Python. In this simulation, we plot the fringes.

Characteristics of the Simulation

- The program asks the user for the values of d (in mm), λ (in nm), D (in m), and ψ (in degrees).
- The libraries used are NumPy and Matplotlib.
- The colour of the fringe curves depend on the wavelength λ given as input.
- The simulator only plots the bright fringes.
- The experiment is in an idealised condition such that all the bright fringes (maxima) have equal intensity.

For example for

$$\begin{aligned} d &= 1\text{mm} \\ \lambda &= 600\text{nm} \\ D &= 1\text{m} \\ \psi &= 80^\circ \end{aligned}$$

we get fringes like this:

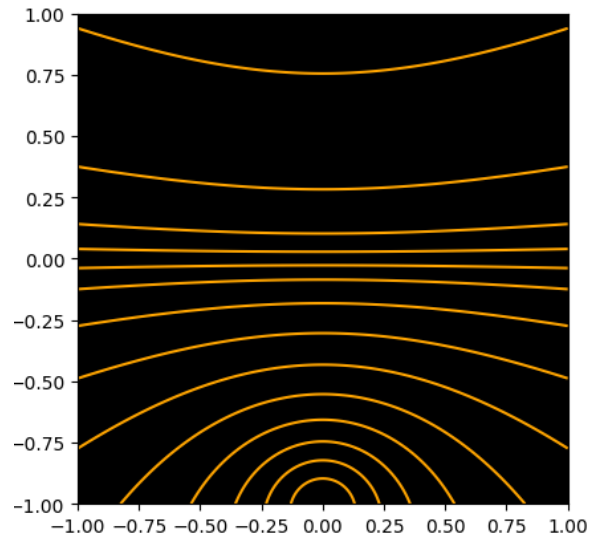


Figure 5: Plotted fringes for the given parameters

The notebook containing the code for this submission can be accessed at [this link](#).

6 References

1. [Double Slit Experiment: Wikipedia](#)
2. SL Loney: The Elements of Coordinate Geometry, Part 1
3. [Desmos 2D Graphing Calculator](#)